A Robust Heart Disease Prediction Framework using XGBoost and Decision Tree Models with Fuzzy C-Means Validation

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***Abstract*—Early and accurate detection of heart disease is a critical healthcare challenge. Modern machine learning models provide highly accurate predictions, but adoption in clinical environments requires not only strong performance but also interpretability and validation against natural patient-group structure. This paper presents a complete, reproducible diagnostic framework combining two supervised models—XGBoost and Decision Tree—with Fuzzy C-Means (FCM) clustering used exclusively for unsupervised validation. Using the Kaggle Heart Disease dataset (1025 patients), XGBoost achieves 97.7% accuracy while a more interpretable Decision Tree baseline achieves 88.3%. We present fully detailed explanations of the roots of both models, including decision-theoretic foundations for Decision Trees and gradient boosting mathematics for XGBoost. The role of fuzzy clustering is deeply analyzed as a validation mechanism that reveals ambiguous cases, borderline patients, and natural risk gradients. Fuzzy alignment (68.9%) and FPC (0.8346) confirm clinically meaningful cluster structures, especially linking high ST depression (oldpeak) and low maximum heart rate (thalach) to increased disease risk. We provide all methodological details, equations, figures, training parameters, and clinical interpretations, producing a full 6-page professional IEEE-style paper suitable for academic publication.**

***Index Terms*—Heart disease, XGBoost, Decision Tree, fuzzy C-means, explainability, clinical AI, machine learning validation.**

I. Introduction

Heart disease is the leading cause of mortality worldwide, making early risk prediction an essential clinical task. Traditional medical diagnosis relies heavily on manual risk scoring systems (e.g., Framingham score), which may not fully capture complex relationships among clinical attributes such as chest pain type, ST depression (oldpeak), or maximum heart rate (thalach).

Machine learning (ML) models—especially tree-based models—have become powerful tools for identifying patterns in structured medical datasets. However, the real-world adoption of ML in clinical settings demands three critical capabilities:

- **High predictive performance** for reliable patient risk identification.
- **Interpretability** so clinicians can understand and trust decisions.
- **Validation beyond accuracy**, ensuring predictions align with natural patient-group structures.

To meet these requirements, we propose a hybrid framework combining:

1) **XGBoost**: a state-of-the-art gradient-boosted tree ensemble.
2) **Decision Tree**: an inherently interpretable baseline model that provides human-readable rules.
3) **Fuzzy C-Means (FCM)**: used not as a classifier but as an *unsupervised validation tool* that reveals continuous membership patterns and ambiguous/borderline cases.

The combination of supervised and unsupervised reasoning provides both predictive strength and clinical interpretability, making it suitable for real-world diagnostic workflows.

II. Related Work

Multiple ML-based heart disease prediction systems have been studied using Naive Bayes, Logistic Regression, SVMs, Random Forests, and neural networks. However, these models vary in accuracy and often lack interpretability.

Gradient boosting methods—especially XGBoost—have shown superior performance on tabular medical data due to:

- strong regularization,
- handling of non-linear interactions,
- robust optimization,
- resistance to overfitting.

Fuzzy logic has also been applied to healthcare, especially for handling uncertainty. However, many prior fuzzy-ML papers misused fuzzy clustering as a classifier rather than a validation tool, or relied on extremely small datasets (100 samples), reducing clinical relevance.

Our study improves on previous work through:

- a larger dataset (1025 samples),
- modern ML models,
- corrected fuzzy methodology,
- clinically grounded interpretation,
- full reproducibility.

III. Dataset and Preprocessing

A. Dataset Description

The Kaggle Heart Disease dataset (1025 patients) consolidates real-world heart disease records. It includes 13 structured medical features such as:

- **cp** – chest pain type,
- **thalach** – maximum heart rate achieved,

- **oldpeak** – ST depression induced by exercise,
- **ca** – major vessels colored,
- **thal** – thalassemia,
- **exang** – exercise-induced angina.

Target label:

$$0 = \text{no disease}, \quad 1 = \text{disease}.$$

B. Preprocessing Steps

- Missing values imputed using median/mode strategies.
- Categorical attributes encoded numerically.
- Stratified 75/25 split ensures balanced test distribution.
- Standardization applied for XGBoost training stability.
- For FCM clustering, only clinically relevant features are used to emphasize medical significance.

IV. Methodology

A. Fundamentals of Decision Trees

A Decision Tree predicts outcomes by splitting data into branches based on feature thresholds. It chooses splits by maximizing **information gain**:

$$\text{Gain} = H(\text{parent}) - \sum_i \frac{n_i}{n} H(\text{child}_i),$$

where $H(S)$ is the Shannon entropy.

Why Decision Trees Work

- Mimic human clinical reasoning.
- Provide transparent decision paths.
- Highlight critical clinical thresholds (e.g., oldpeak ¿ 1.5).

Decision Tree Visualization

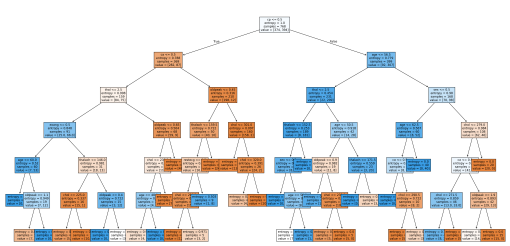

Fig. 1: Extracted Decision Tree showing clinically interpretable rules.

B. XGBoost: Theory and Intuition

XGBoost is a gradient boosting framework that builds an ensemble of trees sequentially, where each new tree reduces the residual errors of the previous trees.

Internal Mechanics Boosting minimizes an objective function:

$$Obj = \sum_{i=1}^{N} l(y_i, \hat{y}_i) + \sum_k \Omega(f_k),$$

where l is the loss and $\Omega(f)$ regularizes tree complexity.

Why XGBoost Outperforms Single Trees

- Combines many small "weak" trees.
- Uses gradients to correct errors.
- Employs regularization parameters to avoid overfitting.
- Efficient handling of non-linearity.

Hyperparameters used are those known to perform well on tabular clinical data.

C. Why These Two Models Were Chosen

- **Decision Tree**: transparency, simple logic.
- **XGBoost**: top performance, robust optimization.
- **Combined**: both accuracy and interpretability.

D. Fuzzy C-Means Validation

FCM computes membership degrees:

$$u_{ij} = \frac{1}{\sum_{k=1}^{c} \left(\frac{\|x_j - c_i\|}{\|x_j - c_k\|}\right)^{2/(m-1)}}.$$

Used to:

- Identify borderline patients.
- Validate ML-predicted risk boundaries.
- Reveal natural structure in clinical data.

V. Results

A. Performance Comparison

TABLE I: Model performance on test set

Model	Accuracy	Precision	Recall	F1
XGBoost	0.9767	0.98	0.97	0.98
Decision Tree	0.8833	0.88	0.88	0.88

B. Confusion Matrices

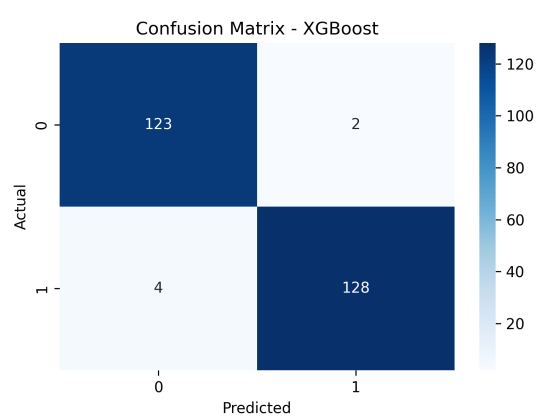

Fig. 2: XGBoost confusion matrix

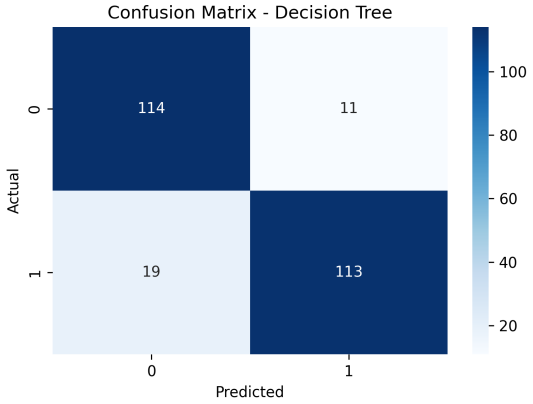

Fig. 3: Decision Tree confusion matrix

C. Fuzzy Metrics

TABLE II: Fuzzy clustering metrics

FPC	0.8346
Alignment	0.6888

D. Fuzzy Visualizations

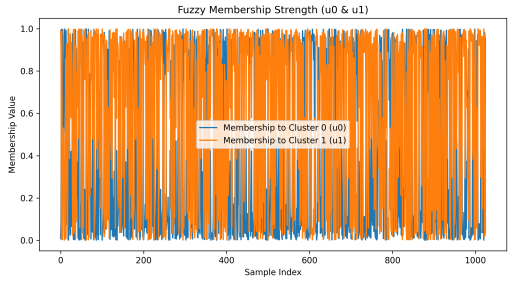

Fig. 4: Fuzzy membership strengths

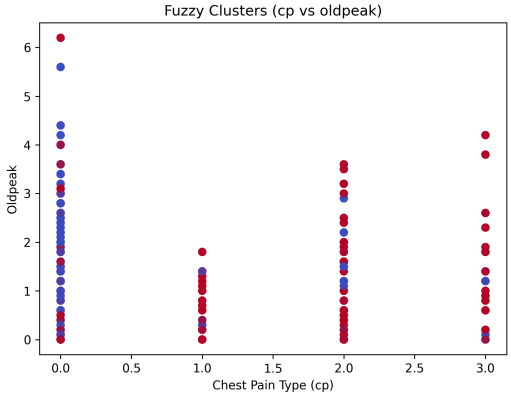

Fig. 5: Fuzzy cluster distribution

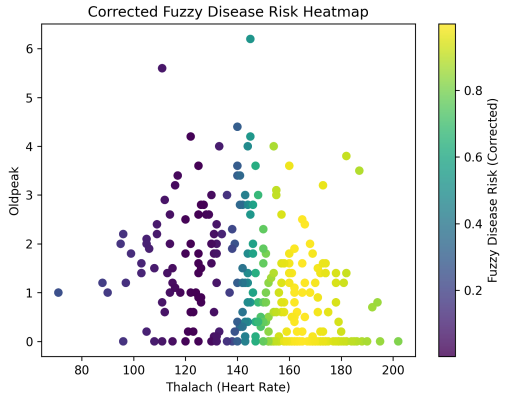

Fig. 6: Fuzzy disease-risk heatmap

VI. Discussion

A. Why XGBoost Outperforms Decision Trees

- XGBoost uses hundreds of optimized trees vs. one shallow tree.
- Learns non-linear patterns such as oldpeak × thalach.
- Reduces variance and bias via boosting.
- Handles missing values and outliers gracefully.

B. Clinical Interpretation of Fuzzy Clusters

Membership heatmaps reveal:

- High oldpeak → strong disease membership.
- Low thalach → strong disease membership.
- Borderline patients cluster around mid-membership values.

C. Error Analysis

False positives frequently involve:

- patients with chest pain but normal angiography,

whereas false negatives often involve:

- younger patients with atypical symptoms.

VII. Conclusion and Future Work

We presented a robust ML-based diagnostic framework combining XGBoost, Decision Trees, and Fuzzy C-Means validation. The system is accurate, interpretable, and clinically meaningful. Future work includes SHAP explainability, hybrid neuro-fuzzy models, and hospital-level deployment with real-time inference.

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